

MUHAMMAD MUDDASIR

+92-3249867842

muddasirrizwan9@gmail.com

muddasirrizwan.com

linkedin.com/in/muddasir-rizwan

EDUCATION

Institute Of Business Administration, Karachi
Bachelors of Science in Computer Science

2021-2025

TECHNICAL SKILLS

Languages: Go (Golang), TypeScript, JavaScript (ES6+), SQL, Python
Backend & Frameworks: Gin, Node.js, Express.js, RESTful APIs, gRPC, Microservices Architecture, Clean Architecture
Tools & Technologies: MongoDB, Docker, Git, AWS, REST APIs, CI/CD (basics), Agile/Scrum

PROFESSIONAL EXPERIENCE

Codet.ai June 2025– Present

Software Engineer

- Architected and maintained high-throughput RESTful microservices using Go (Golang) and the Gin web framework, reducing p99 API latency across core services.
- Designed modular routing, custom middleware pipelines (JWT authentication, structured logging, distributed rate-limiting), and robust error-handling layers in Gin.
- Integrated and optimized persistent data layers with PostgreSQL, implementing connection pooling, migrations, and transactional data integrity.
- Developed asynchronous background workers and concurrent processing pipelines leveraging Go's native goroutines and channels, decoupled via Kafka/RabbitMQ message queues.
- Built the optimizer around a Gaussian-process ensemble (GP-UCB, trust-region leash) with BO-ICL as an advisory LLM surrogate that cannot override hard safety / isolation constraints.

Euronet Feb 2025 – May 2025

Software Engineer

- Developed RESTful APIs using Java, Java EE, for fintech applications. Implemented ISO 8583 standard for payment APIs
- Implemented authentication and role-based access control with JWT.
- Worked with PostgreSQL for database management and query optimization. Integrated third-party payment APIs and optimized backend performance.
- Assisted in debugging and optimizing legacy code to improve system efficiency.

PROJECTS

Crop Yield Predictor | Next.js, Gin, PostgreSQL

- Built a full-stack web application to predict crop yield using geospatial inputs and AI-based models.
- Developed high-performance backend services using Go and Gin to process spatial field coordinates, orchestrate background ML inference workflows, and serve RESTful APIs with minimal latency.
- Utilized PostgreSQL (with PostGIS extensions) and GORM for structured persistence of geospatial boundaries, historical yield predictions, and user metadata.
- Designed a responsive frontend in Next.js integrated with Mapbox GL, enabling users to interactively map agricultural fields and visualize real-time prediction data.

Rulr | Next.js, Gin, Go, Material UI, ReactFlow, Monaco Editor, Redux, React-DnD, Yup

- Designed dynamic JSON rule evaluation and schema validation pipelines in Go, parsing real-time configurations submitted through Monaco Editor to ensure strict data validation before pipeline triggering.
- Developed low-latency REST endpoints to persist workflow configurations, version visual API topologies in PostgreSQL, and validate component state against strong backend data contracts.
- Maintained full-stack synergy by developing intuitive visual workflow builders using ReactFlow, React-DnD, and Monaco Editor paired with strict TypeScript schemas matching backend models.

Askwhy - Filter Bubble | Supabase, Vite, React.js

- Developed AskWhy, a full-stack web application built with Vite and React.js to detect and reduce filter bubbles by analyzing content diversity and recommendation patterns.
- Integrated Supabase for backend services including authentication, real-time database, and scalable cloud storage, enabling secure user management and efficient data handling.
- Designed a modern, responsive UI using Tailwind CSS and shadcn/ui, combining utility-first styling with accessible, reusable components for a clean and consistent user experience.
- Implemented interactive UX features using React Bits animations, smooth transitions, and micro-interactions to enhance engagement.